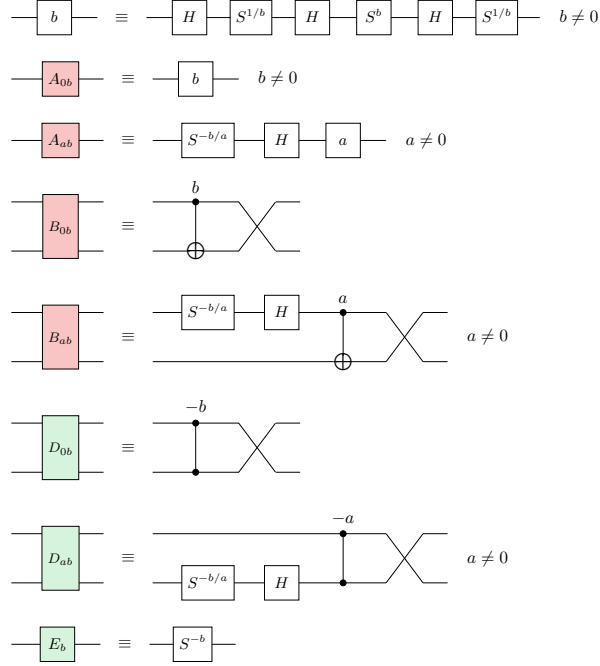


Proof Steps for All Box Relations

For each group, the left column shows **all cases** of the relation (the “what”); the right column shows **representative proof chains** (the “how”). Connectives: $=$ means definitional equality (box expansion/fold); $\equiv_s$ means symplectic equivalence.

Box Definitions



$$E \leftarrow S: S \cdot E[b] \equiv_s E[b-1]$$

$$\boxed{S} \boxed{E_b} \equiv_s \boxed{E_{b-1}}$$

Proof chain (expand $E[b] = S^{-b}$, combine $S \cdot S^{-b} = S^{1-b}$, fold $E[b-1] = S^{-(b-1)}$):

$$\begin{aligned}
 & \boxed{S} \boxed{E_b} \\
 & \parallel \\
 & \boxed{S} \boxed{S^{-b}} \\
 & \parallel \\
 & \boxed{S^{1-b}} \\
 & \parallel \\
 & \boxed{E_{b-1}}
 \end{aligned}$$

$$A \leftarrow H: H \cdot A[\mathbf{v}] \equiv_s \text{dir} \cdot A[\mathbf{v}']$$

$$\text{---} \boxed{H} \text{---} \boxed{A_{0b}} \text{---} \equiv_s \text{---} \boxed{A_{b0}} \text{---} \quad b \neq 0$$

$$\text{---} \boxed{H} \text{---} \boxed{A_{a0}} \text{---} \equiv_s \text{---} \boxed{A_{0,-a}} \text{---} \quad a \neq 0$$

$$\text{---} \boxed{H} \text{---} \boxed{A_{ab}} \text{---} \equiv_s \text{---} \boxed{A_{b,-a}} \text{---} \boxed{S^{1/(ab)}} \text{---} \quad a, b \neq 0$$

Case $b \neq 0, a = 0$: $H \cdot A[0, b] = A[b, 0]$

Expand $A[0, b] = M[b]$, fold as $A[b, 0] = M[b] \cdot H$:

$$\begin{array}{c} \text{---} \boxed{H} \text{---} \boxed{A_{0b}} \text{---} \\ \parallel \\ \text{---} \boxed{H} \text{---} \boxed{b} \text{---} \\ \parallel \\ \text{---} \boxed{A_{b0}} \text{---} \end{array}$$

Case $a \neq 0, b = 0$: $H \cdot A[a, 0] \equiv_s A[0, -a]$

Expand $A[a, 0] = M[a] \cdot H$ (circuit $[H, M[a]]$), apply $H^2 = I$ and symplectic conjugation:

$$\begin{array}{c} \text{---} \boxed{H} \text{---} \boxed{A_{a0}} \text{---} \\ \parallel \\ \text{---} \boxed{H} \text{---} \boxed{H} \text{---} \boxed{a} \text{---} \\ \parallel \\ \text{---} \boxed{A_{0,-a}} \text{---} \end{array}$$

Case $a, b \neq 0$: $H \cdot A[a, b] \equiv_s S^{1/(ab)} \cdot A[b, -a]$

Expand $A[a, b] = M[a] \cdot H \cdot S^{-b/a}$, apply Hadamard conjugation of $S^{-b/a}$:

$$\begin{array}{c} \text{---} \boxed{H} \text{---} \boxed{A_{ab}} \text{---} \\ \parallel \\ \text{---} \boxed{H} \text{---} \boxed{S^{-b/a}} \text{---} \boxed{H} \text{---} \boxed{a} \text{---} \\ \parallel \\ \text{---} \boxed{S^{1/(ab)}} \text{---} \boxed{A_{b,-a}} \text{---} \end{array}$$

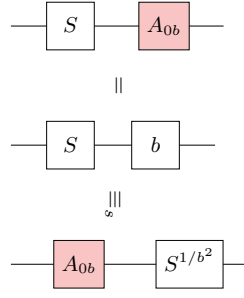
$$A \leftarrow S: S \cdot A[\mathbf{v}] \equiv_s \text{dir} \cdot A[\mathbf{v}']$$

$$\text{---} \boxed{S} \text{---} \boxed{A_{0b}} \text{---} \equiv_s \text{---} \boxed{A_{0b}} \text{---} \boxed{S^{1/b^2}} \text{---} \quad b \neq 0$$

$$\text{---} \boxed{S} \text{---} \boxed{A_{ab}} \text{---} \equiv_s \text{---} \boxed{A_{a,b-a}} \text{---} \quad a \neq 0$$

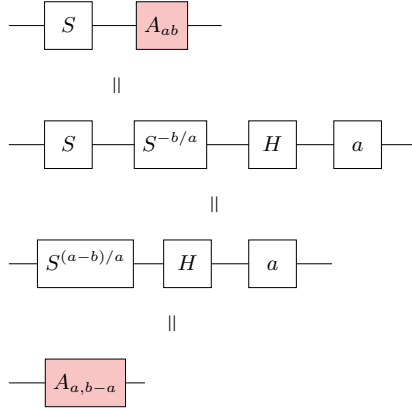
Case $a = 0$: $S \cdot A[0, b] \equiv_s S^{1/b^2} \cdot A[0, b]$

Expand $A[0, b] = M[b]$, use $S \cdot M[b] \equiv_s M[b] \cdot S^{1/b^2}$:

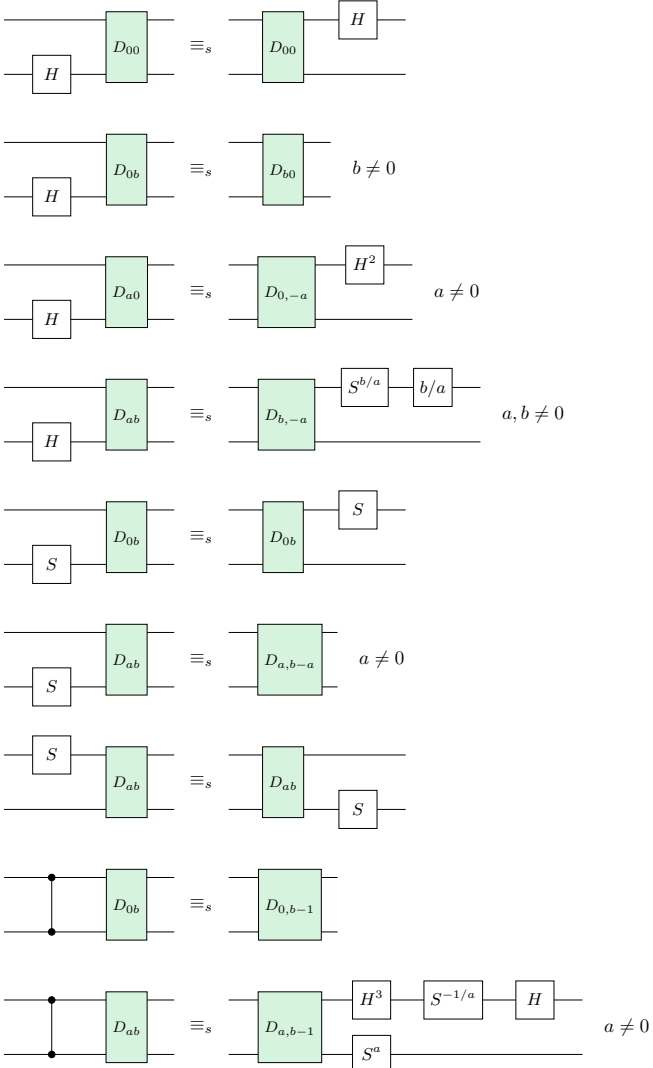


Case $a \neq 0$: $S \cdot A[a, b] = A[a, b - a]$

Expand $A[a, b]$, combine $S \cdot S^{-b/a} = S^{(a-b)/a}$, fold $A[a, b - a]$:

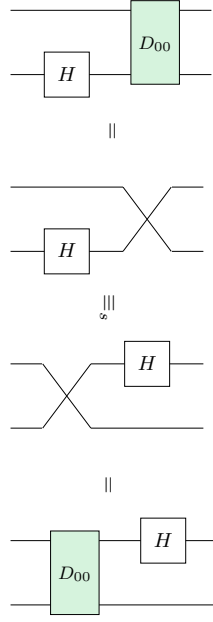


$D \leftarrow H$: $H_{\downarrow} \cdot D[\mathbf{v}] \equiv_s \text{dir} \cdot D[\mathbf{v}']$



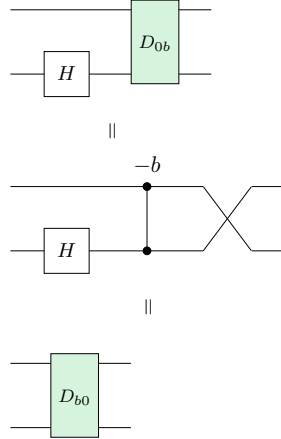
Case $a = b = 0$: $H_{\downarrow} \cdot D[0, 0] \equiv_s D[0, 0] \cdot H_{\uparrow}$

Key axiom: $H_{\downarrow} \cdot \text{Ex} \equiv_s \text{Ex} \cdot H_{\uparrow}$ (Hadamard slides through swap):

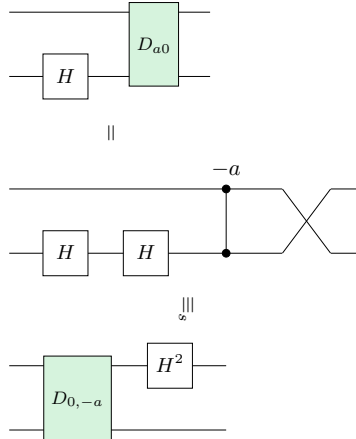


Case $a = 0, b \neq 0$: $H_{\downarrow} \cdot D[0, b] = D[b, 0]$

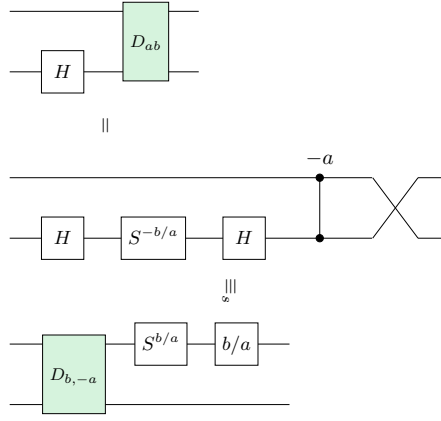
Expand $D[0, b] = [CZ^{-b}, \text{Ex}]$; the result $[H, CZ^{-b}, \text{Ex}]$ is exactly $D[b, 0]$:



Case $a \neq 0, b = 0$: $H_{\downarrow} \cdot D[a, 0] \equiv_s H_{\uparrow}^2 \cdot D[0, -a]$

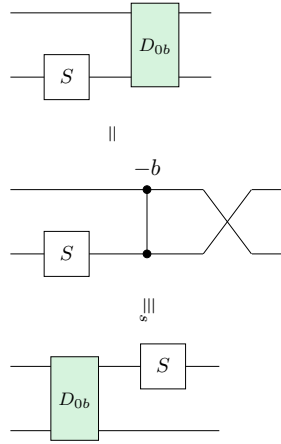


Case $a, b \neq 0$: $H_{\downarrow} \cdot D[a, b] \equiv_s M_{\uparrow}[b/a] \cdot S_{\uparrow}^{b/a} \cdot D[b, -a]$



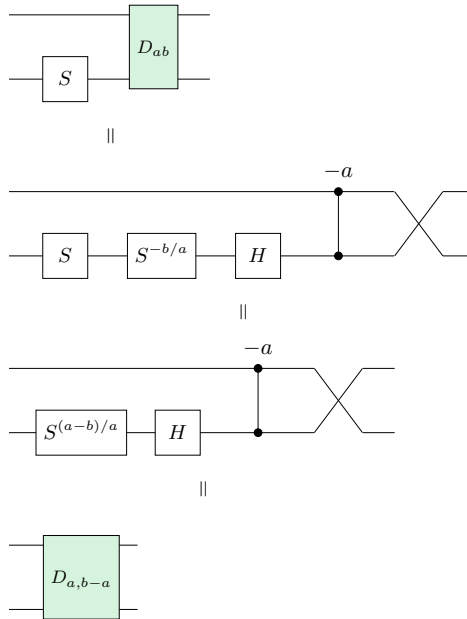
$D \leftarrow S_{\downarrow}$: $S_{\downarrow} \cdot D[\mathbf{v}] \equiv_s \text{dir} \cdot D[\mathbf{v}']$

Case $a = 0$: $S_{\downarrow} \cdot D[0, b] \equiv_s D[0, b] \cdot S_{\uparrow}$

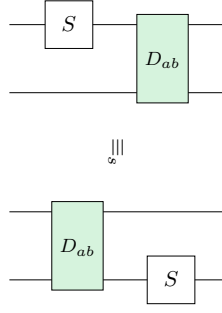


Case $a \neq 0$: $S_{\downarrow} \cdot D[a, b] = D[a, b - a]$

Expand $D[a, b]$, combine $S \cdot S^{-b/a} = S^{(a-b)/a}$, fold $D[a, b - a]$:



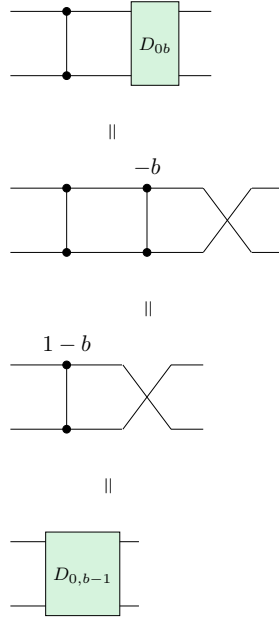
$$D \leftarrow S_{\uparrow}: S_{\uparrow} \cdot D[\mathbf{v}] \equiv_s D[\mathbf{v}] \cdot S_{\downarrow}$$



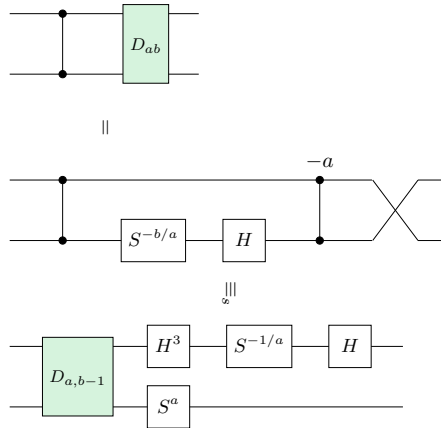
$$D \leftarrow CZ: CZ \cdot D[\mathbf{v}] \equiv_s \text{dir} \cdot D[\mathbf{v}']$$

Case $a = 0$: $CZ \cdot D[0, b] = D[0, b - 1]$

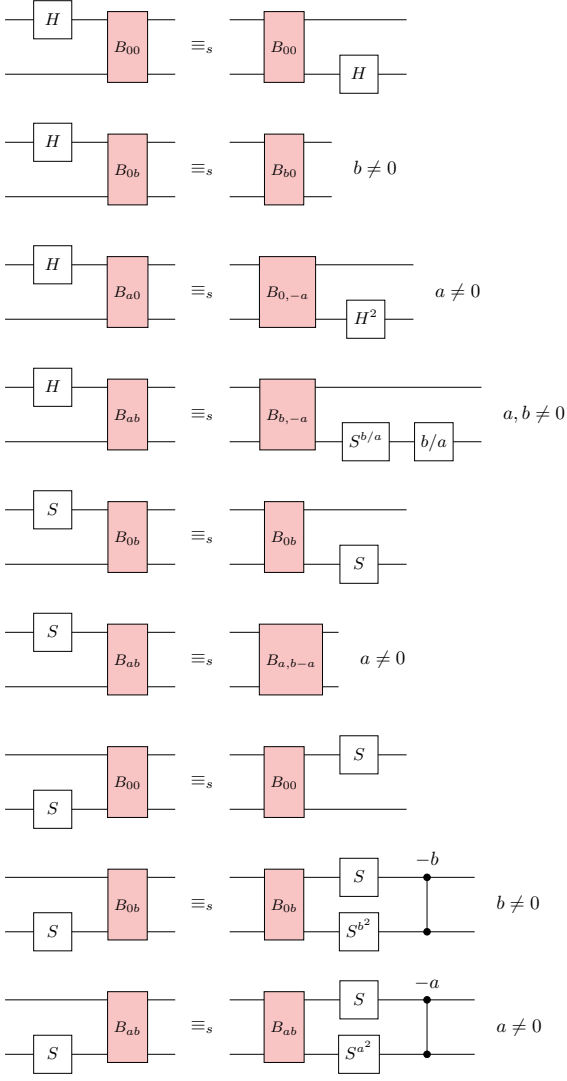
Key step: $CZ \cdot CZ^{-b} = CZ^{1-b}$.



Case $a \neq 0$: $CZ \cdot D[a, b] \equiv_s S_{\downarrow}^a \cdot H_{\uparrow} \cdot S_{\uparrow}^{-1/a} \cdot H_{\uparrow}^3 \cdot D[a, b - 1]$

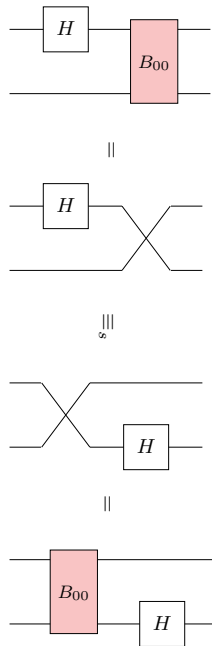


$$B \leftarrow H_{\uparrow}: H_{\uparrow} \cdot B[\mathbf{v}] \equiv_s \text{dir} \cdot B[\mathbf{v}']$$

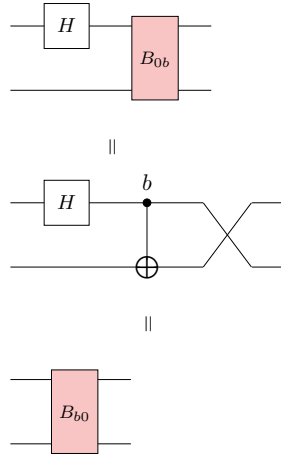


Case $a = b = 0$: $H_{\uparrow} \cdot B[0, 0] \equiv_s B[0, 0] \cdot H_{\downarrow}$

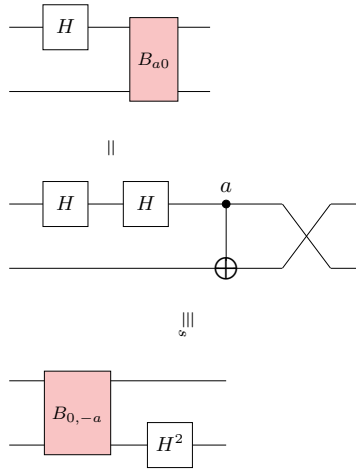
Key: $H_{\uparrow} \cdot \text{Ex} \equiv_s \text{Ex} \cdot H_{\downarrow}$:



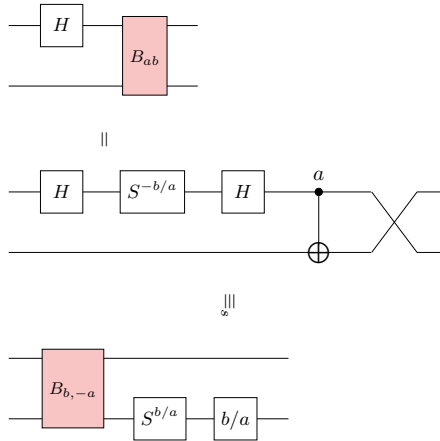
Case $a = 0, b \neq 0$: $H_{\uparrow} \cdot B[0, b] = B[b, 0]$



Case $a \neq 0, b = 0$: $H_{\uparrow} \cdot B[a, 0] \equiv_s H_{\downarrow}^2 \cdot B[0, -a]$

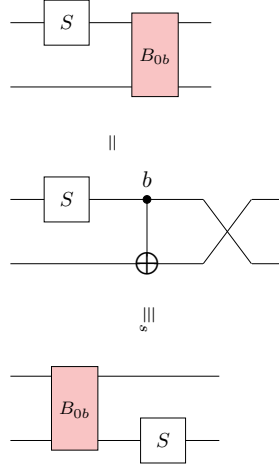


Case $a, b \neq 0$: $H_{\uparrow} \cdot B[a, b] \equiv_s M_{\downarrow}[b/a] \cdot S_{\downarrow}^{b/a} \cdot B[b, -a]$

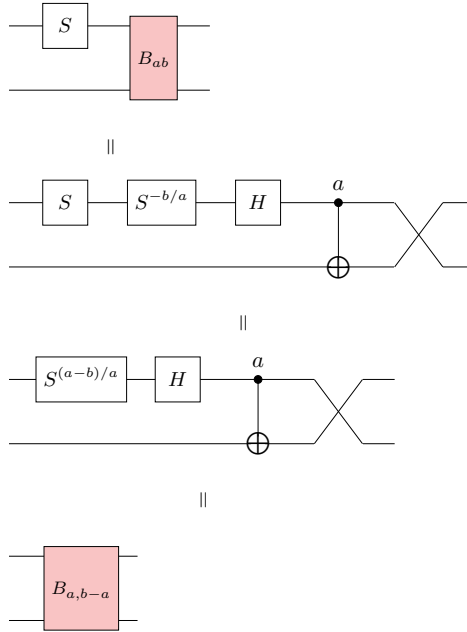


$$B \leftarrow S_{\uparrow}: S_{\uparrow} \cdot B[\mathbf{v}] \equiv_s \text{dir} \cdot B[\mathbf{v}']$$

$$\text{Case } a = 0: S_{\uparrow} \cdot B[0, b] \equiv_s B[0, b] \cdot S_{\downarrow}$$



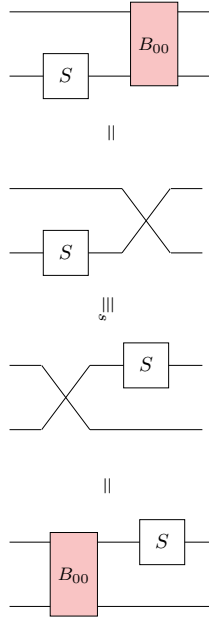
$$\text{Case } a \neq 0: S_{\uparrow} \cdot B[a, b] = B[a, b - a]$$



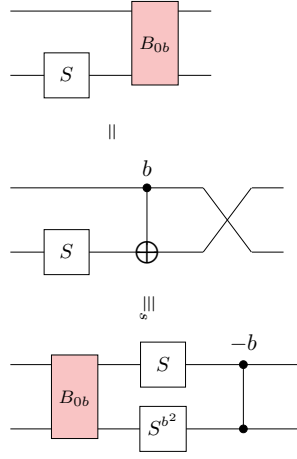
$$B \leftarrow S_{\downarrow}: S_{\downarrow} \cdot B[\mathbf{v}] \equiv_s \text{dir} \cdot B[\mathbf{v}']$$

$$\text{Case } a = b = 0: S_{\downarrow} \cdot B[0, 0] \equiv_s B[0, 0] \cdot S_{\uparrow}$$

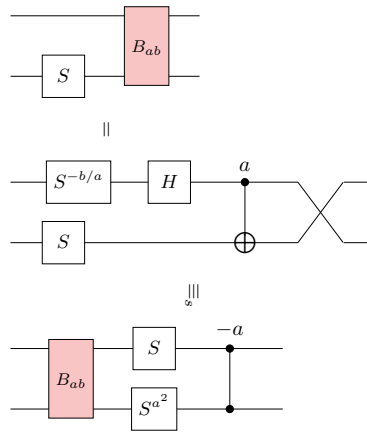
$$\text{Key: } S_{\downarrow} \cdot \text{Ex} \equiv_s \text{Ex} \cdot S_{\uparrow}:$$



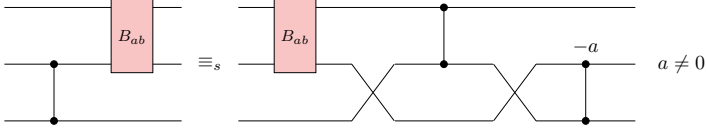
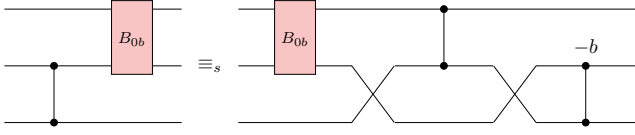
Case $a = 0, b \neq 0$: $S_{\downarrow} \cdot B[0, b] \equiv_s CZ^{-b} \cdot S_{\downarrow}^{b^2} \cdot S_{\uparrow} \cdot B[0, b]$



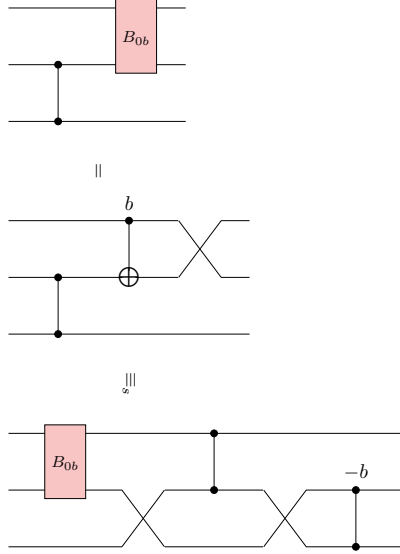
Case $a \neq 0$: $S_{\downarrow} \cdot B[a, b] \equiv_s CZ^{-a} \cdot S_{\downarrow}^{a^2} \cdot S_{\uparrow} \cdot B[a, b]$



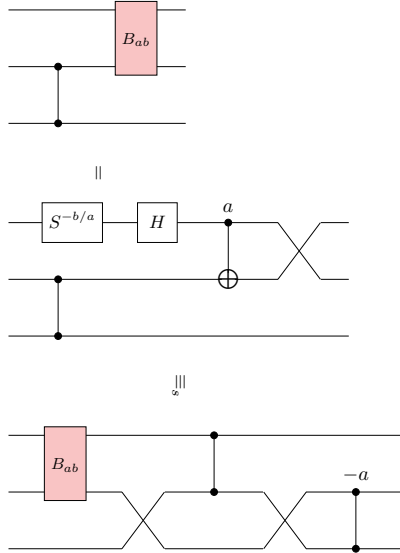
$$B^\uparrow \leftarrow CZ: CZ \cdot B^\uparrow[\mathbf{v}] \equiv_s \text{dir} \cdot B^\uparrow[\mathbf{v}]$$



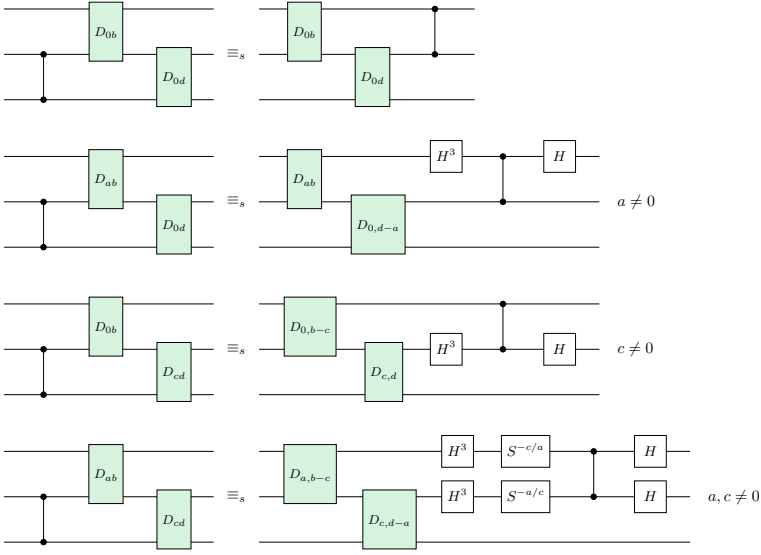
Case $a = 0$: expand $B^\uparrow[0, b]$, apply key axiom



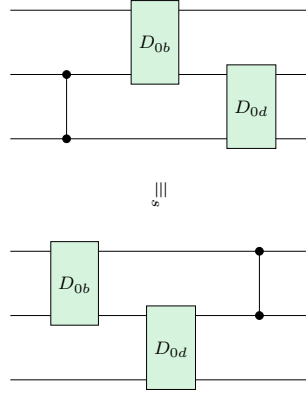
Case $a \neq 0$: expand $B^\uparrow[a, b]$, apply key axiom



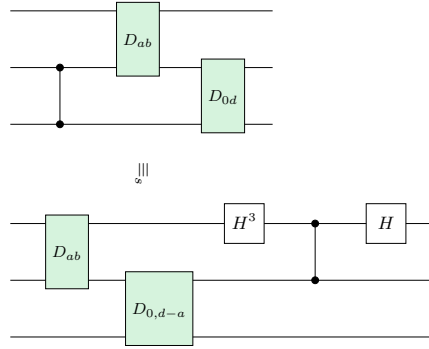
$$DD \leftarrow CZ: CZ \cdot DD[\mathbf{v}_1, \mathbf{v}_2] \equiv_s \text{dir} \cdot DD[\mathbf{v}'_1, \mathbf{v}'_2]$$



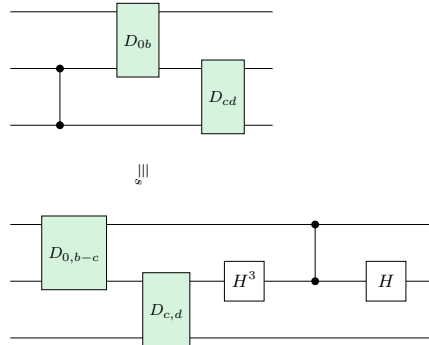
Case 1: $a_1 = a_2 = 0$



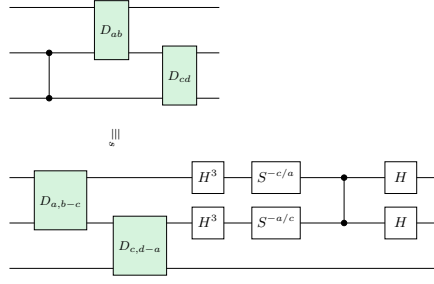
Case 2: $a_1 \neq 0, a_2 = 0$



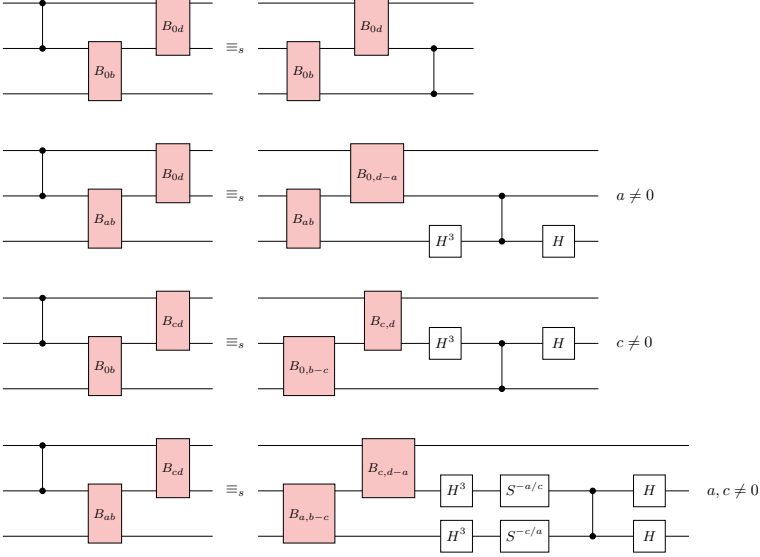
Case 3: $a_1 = 0, a_2 \neq 0$



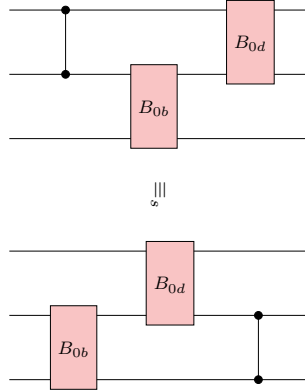
Case 4: $a_1, a_2 \neq 0$



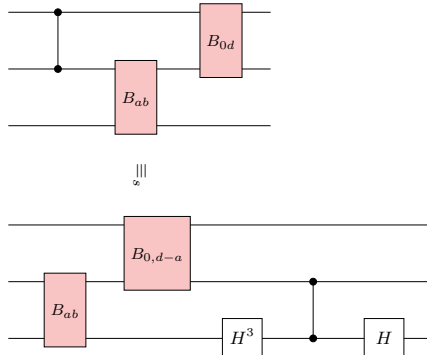
$BB \leftarrow CZ^\dagger: CZ^\dagger \cdot BB[\mathbf{v}_1, \mathbf{v}_2] \equiv_s \text{dir} \cdot BB[\mathbf{v}'_1, \mathbf{v}'_2]$



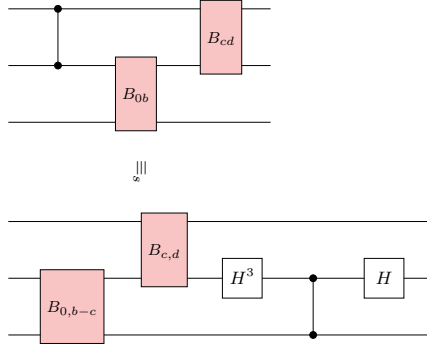
Case 1: $a_1 = a_2 = 0$



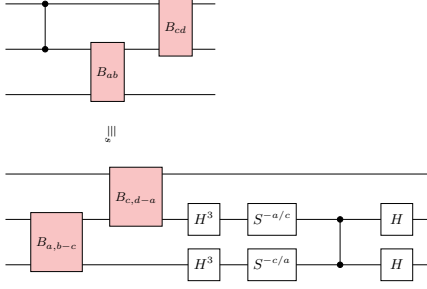
Case 2: $a_1 \neq 0, a_2 = 0$



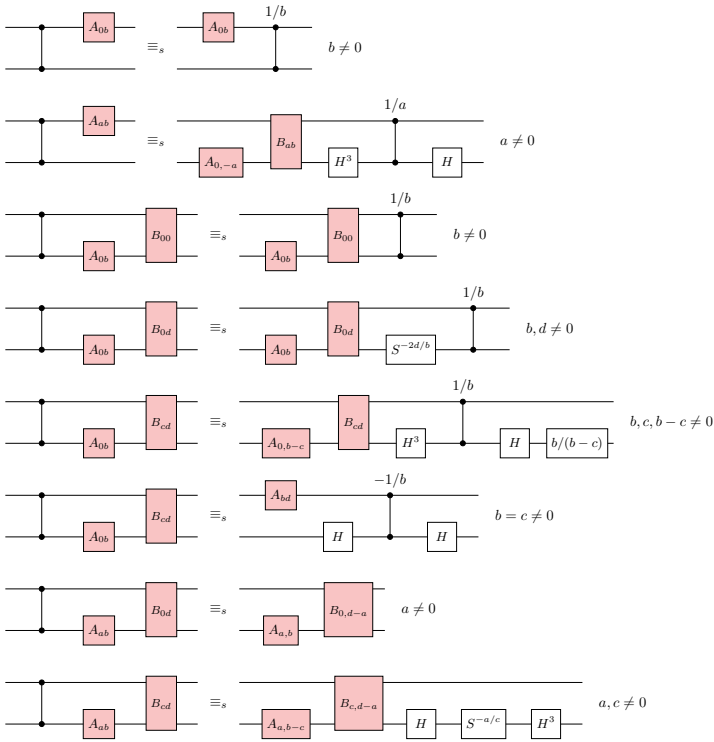
Case 3: $a_1 = 0, a_2 \neq 0$



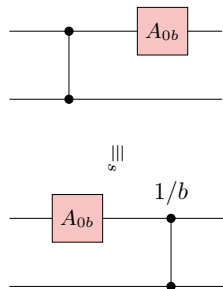
Case 4: $a_1, a_2 \neq 0$



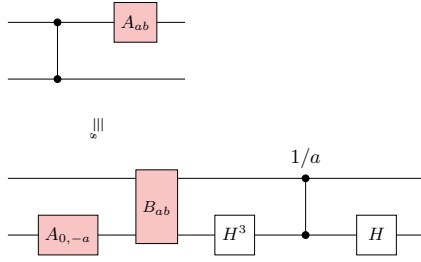
$L \leftarrow CZ: CZ \cdot L[\mathbf{v}] \equiv_s \text{dir} \cdot L[\mathbf{v}']$



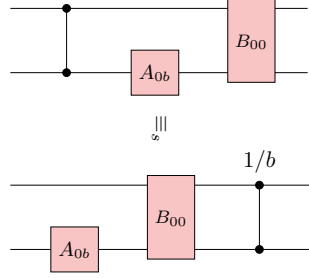
Case 1: $CZ \cdot A^\dagger[0, b] \quad (b \neq 0)$



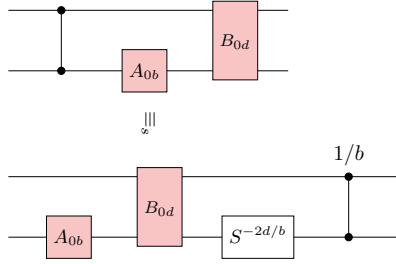
Case 2: $CZ \cdot A^\dagger[a, b] \quad (a \neq 0)$



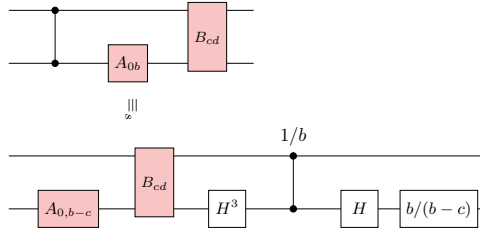
Case 3: $CZ \cdot A[0, b] \cdot B[0, 0]$



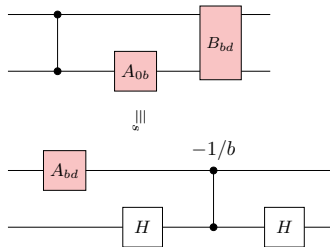
Case 4: $CZ \cdot A[0, b] \cdot B[0, d] \quad (b, d \neq 0)$



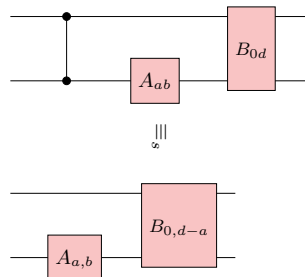
Case 5: $CZ \cdot A[0, b] \cdot B[c, d] \quad (b \neq c)$



Case 6: $CZ \cdot A[0, b] \cdot B[b, d] \quad (b = c)$



Case 7: $CZ \cdot A[a, b] \cdot B[0, d] \quad (a \neq 0)$



Case 8: $CZ \cdot A[a, b] \cdot B[c, d]$ ($a, c \neq 0$)

